

‘**mother nature**’, gradually led to the disintegration of nature. To overcome this problem, it is time that humans act and save the nature for the future generations. The knowledge of organisms, environment, and the interrelationships is vital for the conservation of the Nature.

8.1 ORGANISMS AND ENVIRONMENT

8.1.1 Levels of organization

8.1.2 Environment

8.1.3 Habitat

8.1.4 Niche

Ecology at the organism level is essentially **physiological ecology** which tries to understand how different organisms are adapted to their environments in terms of not only survival but also reproduction. You may have learnt in earlier classes how the rotation of our planet around the

Sun and the tilt of its axis cause annual variations in the intensity and duration of light and temperature, resulting in the occurrence of distinct seasons. These variations together with annual variations in precipitation (rainfall) account for the formation of the major ‘**biomes**’ such as **desert**, **rain forest** and **tundra**.

Regional and local variations within each biome lead to the formation of a wide variety of habitats. On the planet Earth, life exists not just in a few favourable habitats only but even in extreme and harsh habitats such as the hot Rajasthan desert, rain-soaked Meghalaya forests, deep ocean trenches, fast flowing streams, polar regions, high mountain tops, hot thermal springs, and stinking ‘compost pits’, to name a few. Even our intestine is a **unique habitat** for hundreds of species of microbes.

What are the key elements that lead to so much variation in the physical and chemical conditions of different habitats?

The most important ones are temperature, water, light and soil. We must remember that the physico-chemical (**abiotic**) components alone do not characterize the habitat of an organism completely; the habitat includes **biotic components** also - pathogens, parasites, predators and competitors - of the organism with which they interact constantly. We assume that over a period of time, organisms evolved adaptations for survival and reproduction, in their habitats through **Natural Selection**.

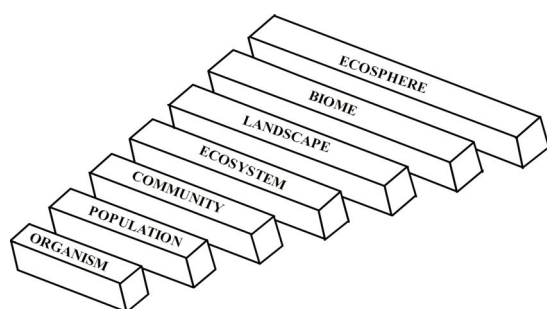


Fig. 8.1 Levels of organization

8.1.1 Levels of organization

Ecology is basically concerned with four levels of biological organization-**organisms**, **populations**, **communities** and **biomes**. In this chapter we explore ecology at organism and population levels.

Ecological Hierarchy means arrangement into a '**graded series**'. Ecological organization consists of eleven integrative levels, ranging from Cell to Ecosphere - cell, tissue, organ, organ- system, organism, population, community, ecosystem, landscape, biome and ecosphere (also called Biosphere).

1. Population

Population is a group of organisms of the same species, living in a specific area at a specific time. For example, the fish belonging to the species *Catla catla* living at a given time, in a pond constitute a 'population'. In the same way, the people (*Homo sapiens*) of India in 2012 form the **Indian population**. Populations are regulated by a set of factors such as **natality**, **mortality**, and **population density**.

2. Community

It is an association of the interacting members of populations of different autotrophic and heterotrophic species in a particular area. In a community, generally one or a few species dominate with reference to their numbers or size. Generally, a community is named after its dominant population. For example, the Himalayan region (at the altitude of 5000 to 8000 ft.) is dominated by the pine group plants – **deodar trees**. So, it is described as the **Pine-Deodar community**.



Fig. 8.2. Pine-deodar community

3. Ecosystem

It is the next level of organization above the level of *biological community*. An **ecosystem** is a functional unit of the biosphere in which members of the community interact among themselves and with the surrounding environment, involving '**flow of energy**' forming a well defined **trophic structure**. An ecosystem can be as small as an aquarium or a tiny puddle. Ecosystems sometimes may overlap, as the boundaries of an ecosystem are flexible.

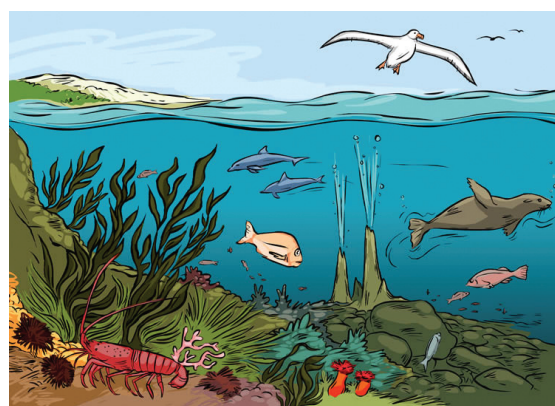


Fig. 8.3 Ecosystem

4. Landscape

It is the unit of land containing different ecosystems (mosaic of ecosystems) surrounded by natural boundaries. It is the level of organization higher than 'ecosystem'.

5. Biome

A 'biome' is a **large community of plants and animals** that occupies a vast **region**. There are 'terrestrial biomes' and 'aquatic biomes'. Terrestrial biomes are characterized by **specific climate** and **dominant vegetation** e.g. tropical rain forest, desert, coniferous forest, tundra etc. Aquatic biomes include freshwater biomes and marine biomes.

6. Ecosphere (Biosphere)

All the habitable zones on the Earth constitute the ecosphere or biosphere. It is the part of the Earth that supports 'life'. It extends several kilometers above the Earth's surface into the atmosphere and extends several kilometers below the ocean's surface. The biosphere comprises all of the Earth's biomes.

8.1.2 Environment

Environment is the sum total of biotic and abiotic factors present around the organisms influencing them in various ways. Any component of the environment that influences the organism is called **environmental factor**. The environmental factors are of three types i) climatic, ii) topographical, iii) edaphic.

- i) Climate refers to average weather conditions like temperature, humidity, cloud cover, wind velocity, rainfall, etc.
- ii) Topography refers to the surface features (physical features such as elevation / altitude) of a place or region. The topography affects the climatic conditions of the region.
- iii) Edaphic factors are related to the soil, and they influence the life of organisms. The main edaphic factors include soil texture, pH, content of organic matter, water, etc.

8.1.3 Habitat and Medium

Ecologically, **habitat** is the place in which an organism lives. It is comparable to the '**address**' of a person (as mentioned in the introduction page to ecology). For instance, the habitat of fish is a pond, lake, sea etc., the habitat of a lion is forest, the habitat of *Ascaris* is the 'small intestine' of man, and so on. The **water** surrounding the body of a fish is called the **medium** and the medium of a lion is the **air** around its body.

8.1.4 Niche

Within a community, each organism occupies a particular biological role or **niche**. Niche is the **functional role** of an organism in an ecosystem (comparable to the '**profession**' of a person, as mentioned in the introduction). It describes the relational position of a species in its ecosystem. For example, the ecological niche of a rose bush growing in the backyard includes absorbing light, water and nutrients for photosynthesis, releasing oxygen into the atmosphere, providing shelter and food for other organisms (e.g. **aphids**) – **in short the 'rose bush' is a 'producer'**.